

HEALTHCONNECT CLINIC — WEEK 5

CROSS-TRACK COLLABORATION REPORT

Primary track: Data Analytics

Partner track: Data Science

1. Collaboration Objective

The Data Analytics work provides an evidence base for the next stage of the HealthConnect project. Week 5 identifies patterns associated with appointment attendance and no-shows and highlights variables that warrant further investigation.

2. Analytical Evidence Shared

The main evidence includes appointment outcome distribution, appointment type, reminder behaviour and channel, previous no-show history, distance to clinic, waiting time and relationships among numerical variables.

Key observations:

- No-shows represent 48.46% of appointments, compared with 46.28% attended and 5.26% cancelled.
- Follow-up appointments have the highest observed no-show rate at 51.23%.
- No-reminder appointments show the strongest no-show pattern among reminder categories examined.
- Patients with five previous no-shows all missed their current appointment, with a slight upward pattern in no-shows as previous no-shows increased.
- Distance to clinic and waiting time do not appear to meaningfully affect no-show behaviour in the current analysis.
- No meaningful correlation was observed among the numerical variables examined.

3. Value to Data Science

The findings provide a focused basis for subsequent modelling or hypothesis development. Previous no-show history, reminder variables and appointment type appear particularly relevant for further investigation, while distance and waiting time should not be treated as strong explanatory variables based on the current exploratory evidence.

4. Project Impact

The collaboration creates a clear hand-off from descriptive analytics to future predictive or decision-support work. The distinction between association and causation is retained so that exploratory findings are not presented as causal conclusions.

5. Contribution Summary

Data Analytics contributes a quantified view of the attendance problem, evidence on potentially important variables, and explicit documentation of weaker relationships. This helps the next stage of the project focus on the strongest analytical signals.